CSE563 — Unit VI Numerical Worksheet

Algorithmic and Optimization Approaches

40 questions • CO6 • Student Worksheet

Instructions

Attempt all 40 questions. For dynamic programming, **show the full table** — the table carries the marks, and where the question asks *which* items or *which* subsequence, the traceback must be shown too. For backtracking, make the pruned branches visible. Refer to the *Unit VI Numerical Guide* for a worked example of each problem type.

A. Greedy Approach (Q1–Q8)

Q1. Activities with (start, finish): $(1,4), (3,5), (0,6), (5,7), (3,9), (5,9), (6,10), (8,11), (8,12), (2,14), (12,16)$. Apply the greedy rule and give the maximum set of non-overlapping activities.

Q2. State the greedy-choice property and optimal substructure, and explain why sorting by *finish* time (rather than start time) is the correct rule for activity selection.

Q3. Fractional knapsack with capacity $W = 20$ and items (w, v): $A(5, 50)$, $B(10, 60)$, $C(8, 64)$, $D(4, 20)$. Find the maximum value.

Q4. Repeat the previous question as a *0/1* knapsack (items cannot be divided) and compare with the greedy answer.

Q5. Construct Huffman codes for frequencies A:45, B:13, C:12, D:16, E:9, F:5. Give the code lengths, the total bits and the saving over a fixed-length code.

Q6. For coins $\{1, 7, 10\}$ and amount 15, show that the greedy algorithm is not optimal.

Q7. Give one problem where greedy is provably optimal and one where it is not, stating in each case which property holds or fails.

Q8. A machine processes jobs with durations $6, 3, 8, 2, 5$. Scheduling to minimise the *average completion time* is greedy. Give the rule, the schedule and the average.

B. Divide and Conquer (Q9–Q14)

Q9. Multiply $x = 5678$ and $y = 1234$ using Karatsuba's method, showing the three products.

Q10. State the recurrences for schoolbook and Karatsuba multiplication and solve both, quantifying the improvement.

Q11. Find the maximum subarray sum of $A = [-2, 1, -3, 4, -1, 2, 1, -5, 4]$ using the divide-and-conquer approach. Show the three cases.

Q12. Write the recurrence for the divide-and-conquer maximum-subarray algorithm, solve it, and compare with Kadane's algorithm.

Q13. State and solve the recurrences for merge sort, binary search and Strassen's matrix multiplication, naming the master-theorem case in each.

Q14. Explain the difference between divide-and-conquer and dynamic programming, and state the property that decides which to use.

C. Dynamic Programming (Q15–Q26)

Q15. Solve the 0/1 knapsack with capacity $W = 7$ and items (w, v): $1(1, 1)$, $2(3, 4)$, $3(4, 5)$, $4(5, 7)$. Give the full table.

Q16. Trace back through the table of the previous question to identify which items are chosen, and verify the weight and value.

Q17. Solve the 0/1 knapsack with $W = 8$ and items (w, v): $1(3, 4)$, $2(4, 5)$, $3(2, 3)$. Give the table, the optimal value and the items.

Q18. Find the LCS of $X =$ `ABCD` and $Y =$ `ACBAD`, giving the table and the traceback.

Q19. Find the LCS length of $X =$ `AGGTAB` and $Y =$ `GXTXAYB`.

Q20. Solve the minimum coin change for coins $\{1, 3, 4\}$ and amount 6, giving the full DP array, and compare with the greedy answer.

Q21. Solve the minimum coin change for coins $\{2, 5, 10\}$ and amount 13, or show it is impossible.

Q22. Compute the Fibonacci number F_{10} by tabulation, and state the complexity of the naive recursive method versus the DP method.

Q23. State the five steps of designing a dynamic-programming solution, and apply them to the rod-cutting problem informally.

Q24. Explain the difference between memoisation and tabulation, with the advantages of each.

Q25. Name four classic dynamic-programming problems and give the state and recurrence for each.

Q26. Explain why the 0/1 knapsack DP running in $\Theta(nW)$ does not prove that knapsack is in P.

D. Backtracking (Q27–Q32)

Q27. State the two conditions checked when placing a queen in the N-Queens problem, and explain why no row check is needed.

Q28. Trace the backtracking search for 4-Queens starting from column 0 in row 0, showing where pruning occurs.

Q29. Give both solutions to the 4-Queens problem and state the relationship between them.

Q30. Find all subsets of $\{2, 4, 6, 8\}$ summing to 10 by backtracking, showing the pruning.

Q31. Explain the difference between backtracking and branch and bound, naming a problem suited to each.

Q32. Name three problems solved by backtracking and describe the pruning condition used in each.

E. Genetic and Nature-Inspired Algorithms (Q33–Q40)

Q33. A GA maximises $f(x) = x^2$ on $x \in [0, 31]$ with 5-bit chromosomes $P_1 = 01000$, $P_2 = 11011$, $P_3 = 00111$, $P_4 = 10100$. Decode each and compute the fitness, selection probability and expected copies.

Q34. Using the mating pool $\{P_2, P_2, P_4, P_1\}$ from the previous question, perform single-point crossover on (P_2, P_4) at position 2 and on (P_2, P_1) at position 3. Decode the four children.

Q35. Apply a mutation flipping the second bit of $C_1 = 11100$ from the previous question. Give the new value and fitness, and comment on whether the mutation was beneficial.

Q36. A population has fitnesses $500, 8, 6, 4$. Compute the selection probabilities, diagnose the problem and give three remedies.

Q37. Compute the simulated-annealing acceptance probability for $\Delta = +6$ at $T = 12$, $T = 3$ and $T = 1$. *Use* $e^{-0.5} = 0.6065$, $e^{-2} = 0.1353$, $e^{-6} = 0.00248$.

Q38. A simulated-annealing run starts at $T_0 = 100$ with cooling factor $\alpha = 0.9$. Find the temperature after 5 steps, and how many steps are needed to fall below $T = 1$.

Q39. Perform one PSO velocity and position update for a particle with $x = 4$, $v = -2$, $p = 6$, $g = 10$, $w = 0.5$, $c_1 = c_2 = 2$, $r_1 = 0.6$, $r_2 = 0.4$. Identify which term dominates.

Q40. An ant at city 1 may move to city 2 or city 3, with $\tau_{12} = 4$, $d_{12} = 5$, $\tau_{13} = 1$, $d_{13} = 2$, $\alpha = 1$, $\beta = 2$. Compute both transition probabilities. Then update τ_{13} given $\rho = 0.4$, $Q = 60$ and an ant tour of length 30 using that edge.

Q41. State the No Free Lunch theorem and explain what it implies for choosing between an exact algorithm and a metaheuristic.